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The Emerging Neurobiology of Attention Deficit Hyperactivity Disorder: The Key Role of the Prefrontal Association Cortex

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Abstract

Attention deficit/hyperactivity disorder (ADHD) is characterized by symptoms of inattention, impulsivity, and locomotor hyperactivity. Recent advances in neurobiology, imaging, and genetics have led to a greater understanding of the etiology and treatment of ADHD. Studies have found that ADHD is associated with weaker function and structure of prefrontal cortex (PFC) circuits, especially in the right hemisphere. The prefrontal association cortex plays a crucial role in regulating attention, behavior, and emotion, with the right hemisphere specialized for behavioral inhibition. The PFC is highly dependent on the correct neurochemical environment for proper function: noradrenergic stimulation of postsynaptic alpha-2A adrenoceptors and dopaminergic stimulation of D1 receptors is necessary for optimal prefrontal function. ADHD is associated with genetic changes that weaken catecholamine signaling and, in some patients, with slowed PFC maturation. Effective pharmacologic treatments for ADHD all enhance catecholamine signaling in the PFC and strengthen its

regulation of attention and behavior. Recent animal studies show that therapeutic doses of stimulant medications preferentially increase norepinephrine and, to a lesser extent, dopamine, in the PFC. These doses reduce locomotor activity and improve PFC regulation of attention and behavior through enhanced catecholamine stimulation of alpha-2A and D1 receptors. These findings in animals are consistent with improved PFC function in normal human subjects and, more prominently, in patients with ADHD. Thus, a highly cohesive story is emerging regarding the etiology and treatment of ADHD.

Attention deficit/hyperactivity disorder (ADHD) is characterized by symptoms of inattention, poor impulse control, and increased motor activity.¹ In the last 20 years, advances in the fields of neuroscience and genetics have provided new insights into this common disorder. We have learned how genetic alterations can affect neural circuits and lead to the symptoms of ADHD, and how correcting these alterations can lead to rational treatments. Much of the research on ADHD has pointed to weaknesses in the prefrontal cortex (PFC), the most highly evolved of the association cortices. The PFC regulates attention and behavior through its widespread connections to sensory and motor cortices, and to subcortical structures such as the basal ganglia and cerebellum. Imaging studies have demonstrated that patients with ADHD have alterations in PFC circuits and demonstrate weaker PFC activation while trying to regulate attention and behavior. The PFC requires optimal levels of norepinephrine (NE) and dopamine (DA) for proper functioning. Genetic studies have consistently noted alterations in genes involved in catecholamine transmission in patients with ADHD. All pharmacologic treatments for ADHD strengthen catecholamine signaling in the PFC and ameliorate symptoms. This article provides a brief summary of the neurobiology of ADHD.

OVERVIEW OF THE PREFRONTAL CORTEX

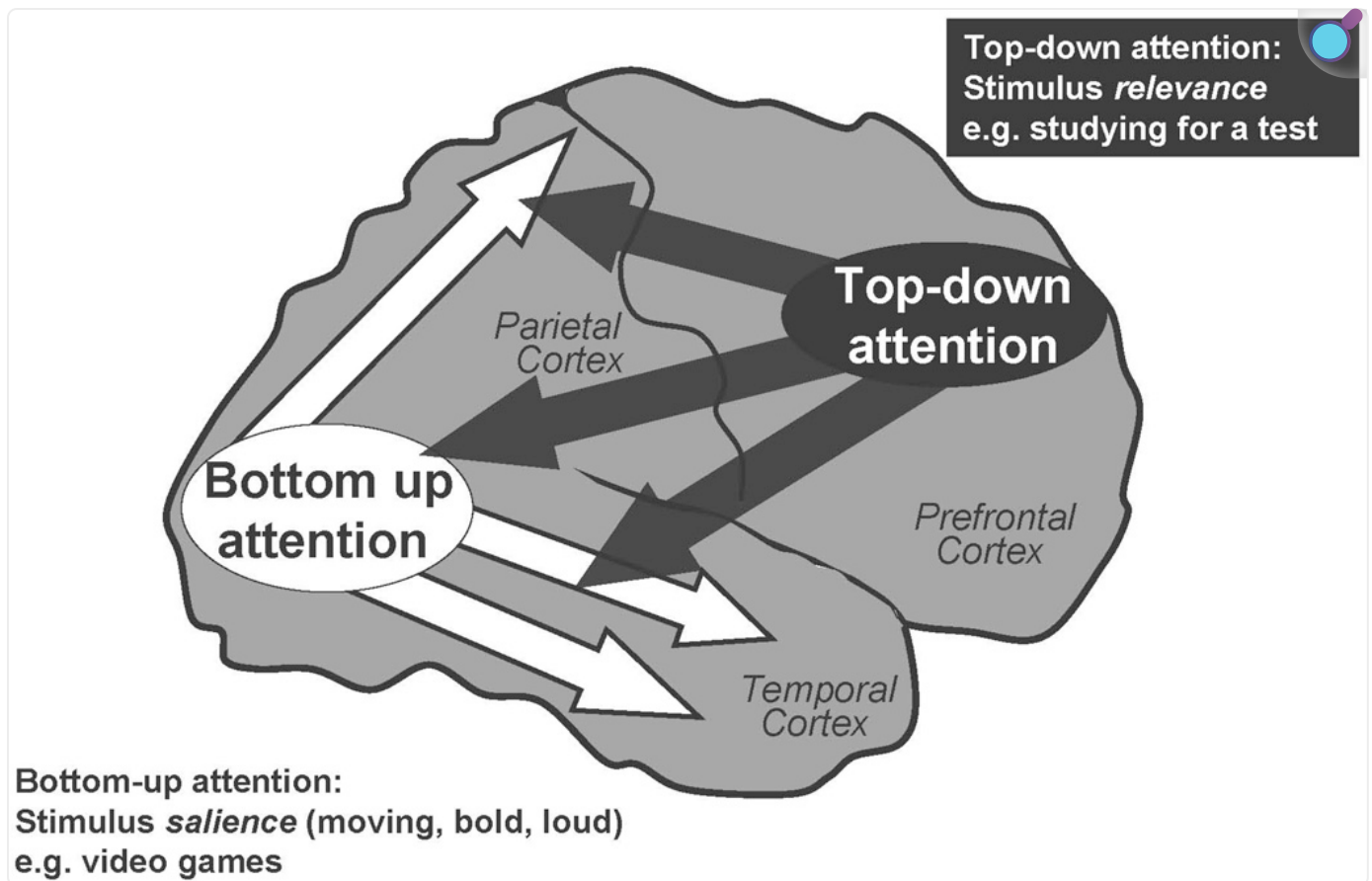
The PFC is highly developed in humans and consists of the cortex anterior to the motor and premotor cortices in the frontal lobe. The functions of the PFC are specialized by region. In right-handed individuals, portions of the left hemisphere are involved with the generation of language (e.g., Broca's area), and the right hemisphere is particularly important for the regulation of attention, behavior, and emotion.² The dorsal and lateral portions of the PFC regulate attention and motor responses, and the ventral and medial portions regulate emotion.^{2, 3} The PFC has extensive connections throughout the brain to orchestrate thoughts and responses⁴ and to provide intelligent decision making, insight, and judgment.^{5, 6} The PFC is essential for the so-called executive functions, allowing us to organize and plan for the future and to inhibit responses to distractions in order to achieve a goal.³ Not surprisingly, the PFC is the brain structure that is last to mature, with full maturation occurring only in late adolescence.⁷⁻⁹ The PFC is also especially sensitive to its neurochemical environment: like

Goldilocks, it needs to have everything “just right” for proper function.¹⁰ Thus, this brain region is particularly vulnerable to environmental and genetic insults.

Regulation of Attention

The PFC mediates “top-down” attention, regulating our attention so that we devote our resources to that which is relevant to our goals and plans.^{11–15} The PFC allows us to concentrate and sustain our attention, especially under “boring conditions” such as long delays between stimuli (e.g., a teacher who talks slowly).¹⁶ The PFC helps us to focus on material that is important but not inherently salient (e.g., studying for a test, reading homework) and to inhibit internal and external distractions.^{17–21} The PFC allows us to divide and shift our attention as appropriate with task demands (so-called multi-tasking)^{2, 22} and to plan and organize for the future.²³ As described above, many of the attentional functions of the PFC are the purview of the right hemisphere, and lesions to this hemisphere induce distractibility and poor concentration.²⁴ The PFC accomplishes top-down attentional regulation through its extensive connections back to the sensory cortices for gating of sensory inputs ([Figure 1](#))^{4, 25} The PFC is able to suppress processing of irrelevant stimuli and enhance the processing of relevant stimuli through these extensive connections.

Figure 1.



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The PFC regulates “top-down” attention, allocating and directing attentional resources based on stimulus relevance. Top-down attention includes stimulus gating, reducing distractibility and sustaining attention on relevant information. These operations are thought to arise from PFC projections back to the sensory cortices. In contrast, the posterior sensory cortices mediate “bottom-up” attention, processing sensory characteristics based on stimulus salience. Most patients with ADHD have difficulties with top-down attention regulation.

Attention problems in children with ADHD are diagnosed using the Inattention scale in the *Diagnostic and Statistical Manual of Mental Disorders (DSM), Fourth Edition*.²⁶ These symptoms of inattention generally refer to problems with top-down attention, as exemplified in children who are easily distracted, have difficulty sustaining attention on “boring” material but are readily captivated by more salient stimuli, e.g., they are able to attend to video games but are not able to listen to their teacher. Most children with ADHD have these problems with attention

regulation. However, there are a few children who are truly unable to *pay* attention (usually diagnosed with attention deficit disorder [ADD], rather than ADHD), and these individuals may have problems with posterior attention systems in the parietal and temporal lobes.

The parietal and temporal sensory cortices mediate “bottom-up” aspects of attention.^{[15](#), [27](#)} These cortical systems process stimuli according to inherent salience (e.g., are the stimuli bold, loud, brightly colored, moving), rather than their relevance. Research over the last 20 years has been particularly successful in discovering how visual stimuli are processed and perceived: the ventral stream through the temporal association cortices evaluates visual features, such as lines and colors, to determine *what* things are.^{[13](#), [28](#), [29](#)} Thus, lesions to the inferior temporal cortices cause agnosias (not knowing what something is).^{[11](#)} In contrast, the dorsal stream culminating in the parietal cortices determines *where* things are, and whether they are moving.^{[30](#), [31](#)} The parietal association cortex is essential for orienting our attention,^{[32](#), [33](#)} with the right hemisphere specialized for orienting attention to parts of visual space, and the left hemisphere marshalling our attention to a point in time, e.g., if we are expecting an important event to occur.^{[34](#)} Lesions to the right parietal cortex induce a striking syndrome known as contralateral neglect, where patients have no conscious experience of stimuli in the left visual field.^{[35](#), [36](#)}

Although most children with ADHD or ADD (attention problems without hyperactivity or impulsivity) have problems with attention consistent with PFC deficits, there are likely some children who have weakness in the parietal or temporal cortices, or both, and truly have difficulties paying attention, e.g., a child who is not engaged even by video games. Unfortunately, the term “inattention” does not distinguish between these scenarios, and the current DSM criteria are not helpful in this regard. It will be important that we create better evaluation scales in the future to discern PFC vs. posterior cortical weakness, as the optimal medications for treating PFC deficits may not be ideal for treating posterior cortical problems.

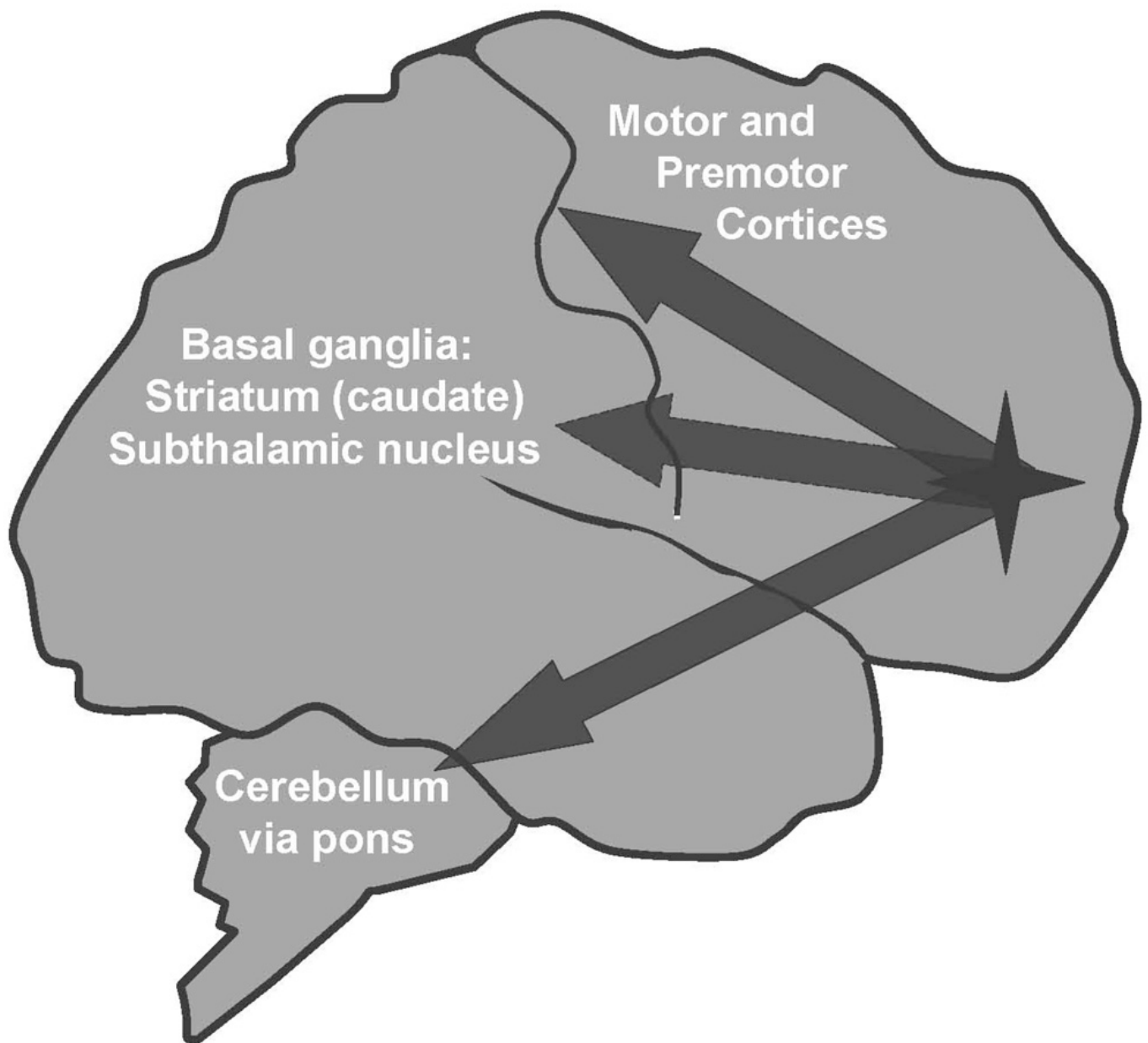
Inhibition of Inappropriate Behaviors

The PFC is also essential for the regulation of behavior, for planning future actions, and for the inhibition of inappropriate responses. For example, lesions to the PFC in monkeys induce locomotor hyperactivity and impulsive responding, similar to what is observed in children with ADHD.^{[37–39](#)} The PFC can guide behavioral output through its massive projections to the motor and premotor cortices, to basal ganglia structures such as the caudate and subthalamic nucleus, and to the cerebellum by way of the pons ([Figure 2](#)).^{[40](#), [41](#)} Thus, lesions in areas such as the caudate or cerebellum can sometimes mimic lesions in the PFC, as they are part of a circuit needed to guide behavioral response. In humans, the right inferior PFC is specialized for behavioral inhibition.^{[42](#)} Functional imaging studies have shown that the right inferior PFC is active when subjects successfully inhibit or stop movements.^{[2](#), [42](#), [43](#)} Conversely, lesions or

weakness to this area impairs the ability to inhibit inappropriate responses.^{[44](#)} A recent study^{[45](#)} showed that manipulations that weaken the right inferior PFC in normal subjects impaired the ability to stop an ongoing motor response (this study used a technology called transcranial magnetic stimulation, where magnetic pulses are directed at the brain to alter the electrical activity of the cortex beneath the skull). As described below, imaging studies have often shown that the right inferior PFC is underactive in patients with ADHD.^{[46](#)}

Figure 2.

The Right Inferior Prefrontal Cortex Is Specialized for Behavioral Inhibition



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The PFC regulates behavior and inhibits inappropriate impulses. In humans, the right inferior PFC is specialized for behavioral inhibition. Projections from this area to the premotor and motor cortices, the basal ganglia (striatum and subthalamic nucleus), and the cerebellum (by way of the pontine nuclei) are likely involved in the inhibition of

inappropriate movements and impulses. In monkeys, blockade of alpha-2A receptors in the PFC induces a pattern of impulsive responding and locomotor hyperactivity.^{68, 69}

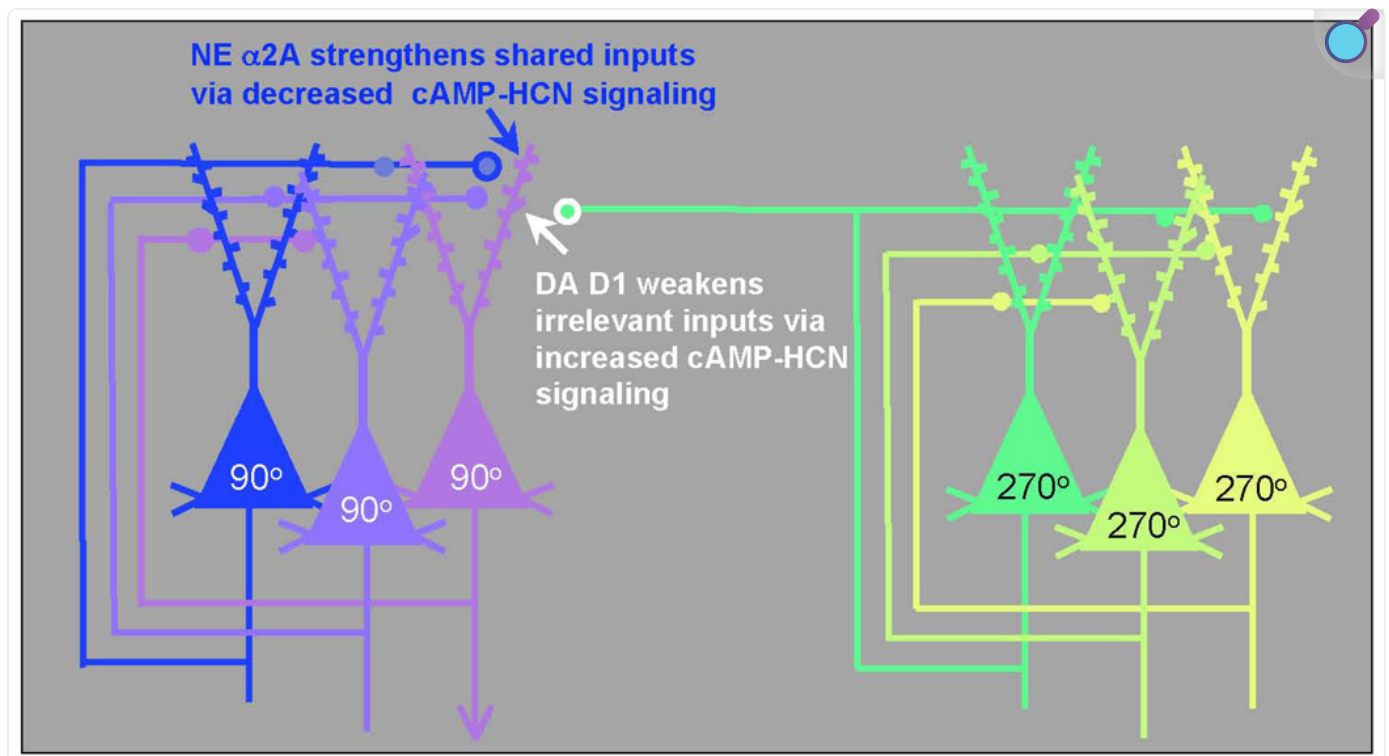
Regulation of Emotion

Whereas the dorsal and lateral portions of the PFC regulate attention and behavior, the ventral and medial portions of the PFC regulate emotion.^{2, 47} The ventral surface of the PFC is often referred to as the orbital cortex, as it sits just above the orbits of the eyes. The ventromedial PFC monitors and inhibits emotions and emotional habits through extensive projections to the amygdala, hypothalamus, and nucleus accumbens, as well as to brainstem nuclei mediating the stress response.⁴⁸⁻⁵¹ Weakness in ventromedial PFC function (especially in the right hemisphere) leads to emotional dysregulation, including disinhibited aggressive impulses.⁵²⁻⁵⁴ Symptoms of aggression and oppositionality (e.g., conduct disorder) are often comorbid with ADHD, particularly in boys.

Neuronal Networks Representing Goals and Rules

The PFC regulates attention, actions, and emotion through networks of PFC neurons. These networks consist of pyramidal cells that use glutamate as their neurotransmitter (schematically illustrated in [Figure 3](#)) and are able to excite each other to maintain firing even in the absence of environmental stimulation.⁵⁵ These networks are able to “keep in mind” information to help guide attention and behavior in a thoughtful manner. For example, they can keep in mind information about where you just left a book you were reading or your reading glasses (e.g., “the book is 90° away from the couch”, as illustrated by the network of 90° cells in [Figure 3](#)). Higher-order networks appear to be able to represent goals and plans for the future (e.g., “Sit in your seat!”, “Do your homework now so you can play tonight”). The neurons in these networks interact with other pyramidal cells through synapses on dendritic spines.⁵⁵ These spines contain NE alpha-2A receptors⁵⁶ or DA D1 receptors,⁵⁷ which dynamically alter the strength of incoming network connections and are essential to PFC function.

Figure 3.



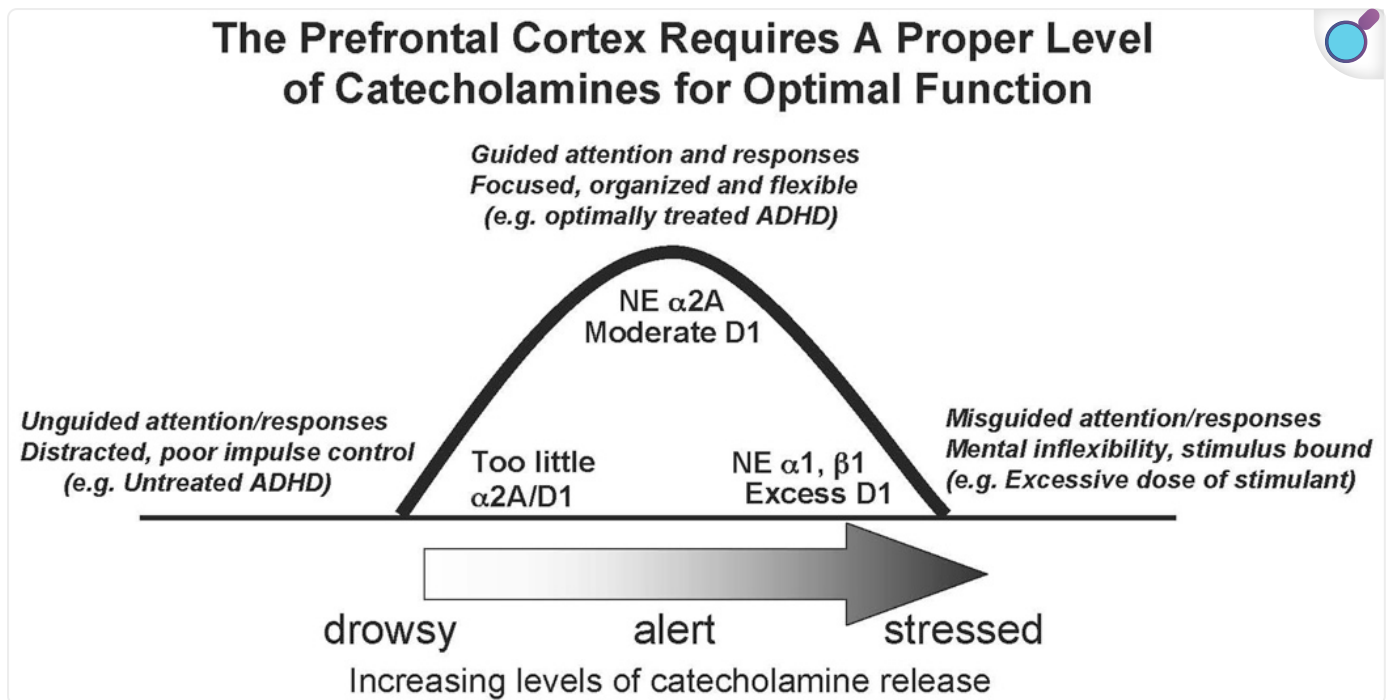
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The PFC guides attention, behavior, and emotion through networks of pyramidal cells. These pyramidal cells engage in recurrent excitation to represent stimuli (e.g., the spatial positions 90° or 270°, as shown here) or to represent goals or rules. The networks interconnect through synapses on dendritic spines that contain NE alpha-2A receptors or DA D1 receptors. Network connectivity is powerfully modulated by the catecholamines: NE alpha-2A receptor stimulation strengthens network inputs from cells with shared network properties by reducing the production of cAMP, thus closing HCN channels and enhancing synaptic inputs to the spine (increasing “signals”). Conversely, optimal levels of DA D1 receptor stimulation weaken irrelevant inputs to the neuron by increasing the production of cAMP, opening HCN channels near the synapse, and shunting incoming information (decreasing “noise”). Thus, for the network representing 90°, alpha-2A receptor stimulation increases the strength of connections from other 90° neurons, and D1 receptor stimulation weakens the connections from neurons with dissimilar characteristics (e.g., 270°). Adapted with permission from Arnsten AF 2007.^{[10](#)}

Optimal Catecholamine Levels Are Needed for Proper Function

NE and DA are important components of the arousal systems that arise from the brainstem and project across the entire cortical mantle, including the PFC.⁵⁸⁻⁶⁰ The PFC requires an optimal level of NE and DA for proper function: either too little (as when we are drowsy or fatigued) or too much (as when we are stressed) markedly impairs PFC regulation of behavior and thought.¹⁰ This is often called the inverted U dose response, as illustrated in [Figure 4](#). Indeed, NE and DA are so critical to PFC function that depleting them is as detrimental as removing the cortex itself.⁶¹ As described below, genetic and imaging studies suggest that many patients with ADHD have inadequate transmission of NE or DA, or both. Treatments for ADHD all enhance NE or DA function, or both. Thus, understanding catecholamine actions in the PFC is essential to our understanding of ADHD. The receptor and intracellular mechanisms by which NE and DA influence PFC networks have now been characterized and are summarized here. In brief, NE stimulation of alpha-2A receptors enhances PFC function by strengthening appropriate network connections (increasing “signals”), and DA stimulation of D1 receptors exerts its beneficial effects by weakening inappropriate connections (decreasing “noise”).¹⁰

Figure 4.



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The regulatory functions of the PFC are highly dependent on its neurochemical state. The catecholamines NE and DA are released based on our state of arousal. Either too little or too much catecholamine release is detrimental to PFC function: there is an inverted U dose-response relationship. Inadequate catecholamine release is associated with fatigue and ADHD, and excessive catecholamine release occurs during uncontrollable stress or very high doses of stimulant medications. NE has its highest affinity for alpha-2A receptors and has lower affinity for alpha-1 and beta-1 receptors. Thus, different receptors are engaged based on the amount of NE released in the PFC. Therapeutic doses of stimulants, atomoxetine, or guanfacine likely normalize catecholamine transmission in patients with inadequate DA or NE levels, or both, thus bringing PFC function to more optimal levels at the top of the inverted U.

Role of Norepinephrine

The beneficial effects of moderate doses of NE occur at postsynaptic alpha-2A receptors on PFC neurons.^{56, 62, 63} Research on alpha-2 actions conducted in the 1970s focused on presynaptic alpha-2 receptors on NE cells and terminals that serve as negative feedback to reduce NE cell firing and NE release.⁶⁴ However, it is now known that the majority of alpha-2 receptors in the

brain are actually postsynaptic to NE cells,⁶⁵ situated, for example, on the dendritic spines of PFC pyramidal cells.⁵⁶ There are 3 subtypes of alpha-2 receptors: the A, B, and C subtypes,⁶⁶ and it is the A subtype that is most important to NE's beneficial actions in the PFC.⁶⁷

NE alpha-2A receptor stimulation improves PFC regulation of attention, behavior, and emotion by strengthening network connections between neurons with shared inputs.⁵⁶ This is illustrated in [Figure 3](#), which shows that stimulation of alpha-2A receptors on the spines of a 90° neuron increases the strength of inputs from other neurons that respond to 90°. Thus, alpha-2A receptor stimulation increases “signals” within PFC networks. Alpha-2A receptor stimulation strengthens network connections by closing “leaky” ion channels near the synapses on dendritic spines. These hyperpolarization-activated cyclic nucleotide-gated (HCN) ion channels pass both sodium and potassium when they are opened by cyclic adenosine monophosphate (cAMP), thus shunting nearby inputs. Stimulation of alpha-2A receptors near the HCN channels stops the production of cAMP, closing the channels and increasing the strength of nearby synaptic inputs.⁵⁶

Stimulation of alpha-2A receptors is essential to PFC function, and blockade of these receptors with yohimbine induces a profile similar to ADHD. In monkeys, infusion of yohimbine directly into the PFC increases locomotor hyperactivity⁶⁸ and impulsivity,⁶⁹ similar to lesions of the same area ([Figure 2](#)). Infusion of yohimbine in to prefrontal cortex also weakens working regulation of memory and attention.⁷⁰ Conversely, stimulation of alpha-2A receptors with guanfacine lessens distractibility and strengthens behavioral regulation.^{56, 67, 71–75} Thus, conditions that lead to inadequate NE stimulation of alpha-2A receptors (including genetic insults in ADHD, as described below) lead to marked PFC dysfunction.

In contrast to the essential effects of moderate levels of NE, high levels of NE, such as those occurring during stress or excessive stimulant doses, impair PFC function.⁷⁶ These detrimental actions occur through engagement of alpha-1 receptors (and possibly beta-1 receptors) that have lower affinity for NE.^{77–79} Stimulation of alpha-1 receptors impairs PFC function by engaging the phosphatidylinositol intracellular signaling pathway,⁸⁰ the pathway that is altered in bipolar disorder.^{81, 82} Overactivation of this pathway suppresses PFC cell firing and markedly impairs PFC function.⁸⁰

Role of Dopamine

As with NE, DA is essential to PFC function.⁸³ DA acts at the D1 family of receptors (D1 and D5) and the D2 family of receptors (D2, D3, D4). Studies of DA actions at the D2 family are just emerging. D2 receptors appear to modulate response-related firing of PFC neurons,⁸⁴ and D4 receptors are concentrated on gamma aminobutyric acid (GABA)-ergic interneurons.⁸⁵ D4

receptor stimulation appears to suppress these inhibitory GABAergic interneurons and thus allow pyramidal neurons to fire.⁸⁶ Genetic weakness in the D4 receptor (e.g., the 7-repeat that is more common in ADHD) should lead to excessive GABAergic inhibition and inadequate activity of PFC pyramidal cells. It is important to note that the D4 receptor can be stimulated by both NE and DA, and that NE has higher affinity for D4 receptors than for adrenoceptors.⁸⁷ Thus, medications that increase NE availability likely influence D4 receptor transmission. However, relatively little research has been done on D4 receptor actions in PFC, and they likely have more complex actions than described here. Instead, most research has focused on the D1 family of receptors, as these are most abundant in the PFC ⁸⁸. Currently, no drugs distinguish D1 from D5 receptors; thus, it should be understood that reference to D1 in this review could apply to actions at either of these receptors.

Moderate levels of DA D1 receptor stimulation improve PFC functions by decreasing “noise”.⁸⁹ D1 receptors appear to be on a different set of spines than alpha-2A receptors; the D1 receptors appear to gate incoming inputs, screening out those that are irrelevant to the present task demands.¹⁰ This is schematically illustrated in [Figure 3](#). D1 receptor stimulation prevents inputs from the 270° neurons from entering the 90° cell. D1 receptors weaken irrelevant inputs to the neuron by increasing the production of cAMP, opening HCN channels near the synapse and shunting the incoming information. Thus, DA and NE have complementary beneficial actions.

However, excessive D1 receptor stimulation (such as occurs during stress) impairs PFC function by weakening too many network connections. Under these conditions, network activity collapses, and responding becomes inflexible.⁸⁹ This may explain the problems with mental flexibility when children take excessive doses of stimulant medication.

ALTERED PREFRONTAL AND CATECHOLAMINE FUNCTION IN ADHD

ADHD and Deficits in PFC Function

Patients with ADHD have symptoms similar to those caused by lesions to the right PFC.^{44, 90–92} Imaging studies have shown reduced size and reduced functional activity of the right PFC in patients with ADHD.^{46, 93–97} Recent studies have also reported more disorganized white matter tracks emanating from the PFC in patients with ADHD, consistent with weaker prefrontal connectivity.^{98, 99} Other brain regions connected to the PFC, e.g., the caudate and cerebellum, have also been reported to be smaller in some studies of children with ADHD.¹⁰⁰ There is also evidence of slower prefrontal maturation in some patients with ADHD.¹⁰¹ However, for many patients, ADHD is a lifelong disorder, as supported by results from imaging studies showing evidence of weakened PFC function and reduced right PFC volume in adults with ADHD

symptoms.^{[102](#), [103](#)} Supporting the notion of ADHD as a highly heritable disorder are imaging studies showing disruptions in prefrontal white matter tracts in both parents and their children when both have ADHD.^{[98](#)}

Genetic Changes in Catecholamine Transmission

As is typical in mental illness, multiple genes contribute a small risk to ADHD symptomology.^{[104](#)} Many studies report alterations in the genes encoding for molecules involved in catecholamine signaling, e.g., the DA D1 and D5 receptors,^{[105-108](#)} the DA and NE transporters,^{[105](#), [108-110](#)} the D4 receptor,^{[106](#), [107](#), [111](#)} the alpha-2A receptor,^{[112-114](#)} and dopamine beta hydroxylase (the enzyme needed for the synthesis of NE).^{[105](#), [115](#), [116](#)} There are also associations with the catabolic enzyme, monoamine oxidase, and some serotonergic genes.^{[104](#)} Recent studies have begun to relate genotype to symptomology. For example, genetic variation in the gene encoding for dopamine beta hydroxylase is related to executive function and the ability to sustain attention.^{[117](#), [118](#)} Thus, patients with two copies of the Taq I polymorphism in ADHD have poorer sustained attention.^{[117](#)} These studies suggest that weaker NE production may impair the PFC circuits mediating the regulation of attention and behavior.

Imaging Studies Show Changes in Catecholamine Transmission

Neuroreceptor imaging also supports weakened catecholamine transmission in ADHD. These studies have all been done in adults with ADHD, given the necessity of using radioactive tracers in positron emission tomography or single photon emission computed tomography. The vast majority of this work has focused on DA mechanisms in the striatum, as there are currently no good tracers to image NE or DA levels in the cortex. There have been mixed results with studies of the DA transporter, with many studies showing increased levels in the striatum,^{[119-121](#)} but other studies found no effect^{[122](#)} or reported decreases,^{[123](#)} possibly reflecting genetic heterogeneity in the DA transporter gene. Recent imaging studies have assessed DA release in the striatum and found evidence of decreased DA release in adult patients with ADHD.^{[124](#)} It is likely that this reflects global reductions in DA release throughout the brain, as earlier studies have suggested reduced catecholamine levels in the PFC as well.^{[125](#)} Reduced DA in the striatum is associated with slowed motor activity, as in Parkinsonism,^{[126](#)} and reduced DA in the PFC produces locomotor hyperactivity in animals.^{[127](#)} Such findings suggest that it is the loss of catecholamines in the PFC that is most important for ADHD symptoms.

ADHD TREATMENTS AND THE NORMALIZATION OF CATECHOLAMINE TRANSMISSION

Therapeutic doses of either stimulant or nonstimulant medications potentiate catecholamine transmission in the PFC. Thus, these agents would normalize catecholamine transmission in patients with genetic abnormalities in these pathways.

Stimulants

The stimulants amphetamine (Adderall® [amphetamine], Vyvanse™ [lisdexamphetamine dimesylate], Shire US Inc., Wayne, Penn.) and methylphenidate (Ritalin® [methylphenidate], Novartis Pharmaceuticals, East Hanover, NJ; Concerta® [methylphenidate extended release], McNeil Pediatrics, Ft. Washington, Penn.) block both catecholamine transporters, the transporter for DA and that for NE. Because there are low levels of DA transporters in the PFC, NE transporters thus clear both NE and DA in this brain region.¹²⁸ Previous biochemical studies of amphetamine and methylphenidate in rodents used excessively high doses that increased locomotor activity, impaired PFC function, and had sensitizing effects on pathways involved with, for example, drug abuse.¹²⁹ Recently, more appropriate, lower doses have been identified which produce blood levels in rats similar to those observed in patients with ADHD who are treated with stimulant medication.^{130, 131} These therapeutic doses of stimulants reduce locomotor activity and improve PFC cognitive function in rats just as they do in humans.^{130–132} Biochemical analyses of these more relevant stimulant doses revealed that they substantially increase both DA and NE release in the PFC but have little effect on catecholamine levels in subcortical areas.¹³¹ These data are consistent with those showing that therapeutic doses of stimulants incur little abuse potential when taken properly. In the rat PFC, therapeutic doses of stimulants increase NE release *more* than they increase DA release,¹³¹ thus, it is inaccurate to refer to these agents as simply dopaminergic. Consistent with dual actions on both NE and DA, the cognitive-enhancing effects of these agents in rodents are blocked by either NE alpha-2 or DA D1 receptor antagonists.¹³² However, higher doses of stimulants impair function of the PFC and induce an inflexible pattern of responding similar to that seen following uncontrollable stress.^{131, 132} These findings with high doses of methylphenidate are likely relevant to the cognitive inflexibility that can occur with excessive doses of stimulant medication.¹³³

Therapeutic doses of stimulants improve PFC functions and enhance the efficiency of PFC activity in normal, young adult subjects.^{134, 135} A similar, but much more pronounced profile is observed in subjects with ADHD.^{136–139} Thus, stimulant actions in ADHD are not paradoxical, but simply more apparent.^{134, 135}

Nonstimulants

Atomoxetine

Atomoxetine (Strattera® [atomoxetine], Eli Lilly, Indianapolis, Ind.) selectively blocks the NE transporter. Administration of atomoxetine increases both NE and DA in the rat PFC,^{[140](#)} indicating the importance of the NE transporter for clearing DA as well as NE in the PFC. Preliminary data indicate that moderate doses of atomoxetine, as with methylphenidate, improve PFC functions through both NE alpha-2 and DA D1 actions, and higher doses can impair PFC function in some animals (Arnsten, unpublished). Recent studies in humans have shown that therapeutic doses of atomoxetine can strengthen response inhibition in normal controls^{[141](#)} as well as in patients with ADHD.^{[142](#)} The therapeutic effects of atomoxetine are consistent with results of previous studies showing that desipramine, a tricyclic antidepressant with high selectivity for the NE transporter, is helpful in treating ADHD-related symptoms, although it has cardiovascular side effects.^{[143](#), [144](#)}

Guanfacine

Guanfacine acts directly at postsynaptic, alpha-2A receptors in the PFC, where it mimics the beneficial effects of NE and strengthens PFC regulation of attention and behavior.^{[56](#)} Animal studies have shown that guanfacine improves a wide range of PFC functions.^{[71](#), [72](#), [74](#), [145–147](#)} As described above, guanfacine improves PFC functions by inhibiting cAMP-HCN channel signaling in dendritic spines, thus strengthening synaptic inputs onto pyramidal neurons and strengthening PFC network connectivity.^{[56](#)} The beneficial effects of guanfacine on PFC function are independent of the drug's sedating actions,^{[71](#), [90](#)} which likely occur at all 3 alpha-2 receptor subtypes (the A, B, and C subtypes). For example, the thalamus is rich in alpha-2B receptors,^{[66](#)} and this structure is key for regulating state of arousal,^{[148](#)} The sedating actions of alpha-2 agonists also likely occur at presynaptic alpha-2A receptors on NE cell bodies and terminals; guanfacine has relatively lower affinity for these presynaptic receptors.^{[149](#)} Guanfacine is currently used in both children and adults with ADHD. It has been shown to improve ratings on both the Inattention and Hyperactivity/Impulsivity scales, consistent with its widespread beneficial effects on many PFC functions.^{[150–152](#)} It is especially helpful in patients who cannot take stimulant medications because of tics, aggressive impulses, or drug abuse liability.^{[150](#)} As with the stimulants, guanfacine also can improve normal subjects,^{[153](#), [154](#)} but it is far more effective in individuals with impaired prefrontal abilities and inadequate catecholamine function.^{[71](#), [90](#)} In view of the fact that it works directly at the receptor to mimic NE, it can be used in subjects having marked catecholamine depletion, as an intact catecholamine system is not required for its actions.

Clonidine

Clonidine has a very rapid onset of action that can be helpful in treating emergent situations. However, it has significant sedative and hypotensive actions that limit its clinical utility.^{[155](#), [156](#)} Clonidine is less selective than guanfacine for the alpha-2A receptor. It has high affinity for the alpha-2B and alpha-2C subtypes as well as the alpha-2A receptor,^{[157](#), [158](#)} and it also has high affinity for imidazoline I1 receptors.^{[159](#)} Clonidine has potent actions at presynaptic alpha-2A receptors, being 10 times more effective than guanfacine at these sites.^{[149](#)} This nonselective profile and potent presynaptic actions likely contribute to clonidine's potent sedating effects. In addition, clonidine's actions at imidazoline I1 receptors in the brainstem are thought to contribute to its marked hypotensive actions.^{[159](#), [160](#)}

In summary, successful pharmacological treatments for ADHD mimic or enhance the beneficial effects of catecholamines on PFC function.^{[161](#)}

DISCUSSION

Over the last 20 years, our understanding of higher cortical function has evolved so that we can now begin to explain the etiology and treatment of ADHD. We have learned that the PFC plays a crucial role in regulating attention, behavior, and emotion. Weaknesses in PFC structure and function, including alterations in catecholamine transmission, likely contribute to the etiology of ADHD symptoms. Effective treatments for ADHD optimize catecholamine signaling in the PFC and normalize PFC regulation of attention and behavior, thus reducing ADHD symptoms.

Abbreviations

ADD

Attention deficit disorder

ADHD

Attention deficit/hyperactivity disorder

cAMP

cyclic Adenosine MonoPhosphate

DA

Dopamine

GABA

Gamma aminobutyric acid

HCN

Hyperpolarization-activated Cyclic Nucleotide-gated

NE

Norepinephrine

PFC

Footnotes

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Author Disclosure

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